Aneeq Hasan Ansari

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Education

NED University of Engineering and Technology Karachi, Pakistan

Bachelor of Engineering in Software Engineering | CGPA: 3.64 *August 2026*

- *Relevant Coursework*: Object-Oriented Programming, Design Patterns, Software Testing (Logic & Graph Coverage, Test Design), Network & Information Security, Artificial Intelligence

Experience

AI and Remote Sensing Intern October 2024 – January 2025

Smart City Lab, National Center for Artificial Intelligence (NCAI) *Karachi, Pakistan*

- Developed a crop yield prediction system for Kharif crops, achieving **99.55% precision** and an **R-squared of 0.93** using Random Forest and SVM models.
- Engineered an automated data extraction pipeline using **Google Earth Engine** to process multi-spectral vegetation indices (NDVI, NDMI, NDRI, MSAVI) from satellite imagery.
- Benchmarked model performance against FAO's **AquaCrop** software, significantly reducing Mean Squared Error (MSE: 0.045) through feature engineering.
- Orchestrated the cleaning and processing of large-scale meteorological and climatic datasets using **Pandas** to identify key yield-limiting factors.
- Authored a Technical Colab Notebook detailing the end-to-end ML workflow from EDA to model evaluation.

Projects

AllerGuard | *React Native, Expo, React, TypeScript, Node.js, Express, Supabase, Post-*GitHub | Sep. 2025 – May. 2026
greSQL, REST APIs, ML Kit

- Built a **Node.js/Express** payment service using the **Strategy design pattern** to route transactions between **Stripe** and **PayFast**, handling webhook notifications, session confirmation, and currency conversion with live status writes to Supabase.
- Designed an allergen prediction pipeline pairing a custom **NLP** parser with **Google Gemini API** calls, including API key rotation across multiple accounts for rate-limit management and a Levenshtein-based fuzzy-matching fallback for offline use.
- Built an offline-first data layer with local **SQLite** caching and background sync to **Supabase**, including cache invalidation logic to keep user allergen profiles consistent across sessions and devices.
- Built the mobile scan flow in **React Native (Expo)** with **expo-camera** and **Google ML Kit** text recognition, including an adjustable bounding-box frame (pan and resize handles) to crop and target specific label regions before extraction.
- Built the **React/TypeScript** admin dashboard's revenue analytics (currency-split gross/admin/doctor revenue, monthly trends, consultation-type and top-doctor breakdowns) and patient profile management views.

Healthcare Zero Trust Security System | *Node.js, Express, JWT, REST APIs,*GitHub | Sep. 2025 – Nov. 2025
Python

- Architected a multi-tier **REST API** backend in Node.js/Express implementing four progressive security layers: Baseline JWT, Zero Trust, Context-Aware, and Privacy-Preserving authentication.
- Engineered middleware pipeline for session management, role-based access control (RBAC), and continuous verification — reducing unauthorized access attempts through dynamic trust score evaluation.
- Implemented field-level **AES-256-GCM encryption**, differential privacy (Laplace mechanism), and a simplified Paillier homomorphic encryption scheme for computation on encrypted patient data.
- Built an automated benchmarking suite (100 requests/tier) and Python visualization pipeline, measuring a **88.6% latency overhead** in the Privacy-Preserving tier vs. the 5ms baseline.

EV Charging Network Management System | *C++, OOP, File I/O* GitHub | Jan. 2025 – May 2025

- Designed a multi-module system using **object-oriented principles** (inheritance, polymorphism, encapsulation) to manage charging stations, sessions, and user accounts.
- Implemented **file-based data persistence** to store and retrieve session records and station states across application restarts, simulating a real-world database layer.

- Built an exception handling framework to manage invalid inputs, session conflicts, and hardware fault scenarios, ensuring robust system behaviour under edge cases.
- Developed a real-time charging progress tracker with dynamic status updates, demonstrating UI-logic separation and state management fundamentals.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), Python, SQL, C++, HTML/CSS
Frontend & Mobile: React.js, React Native, Expo, Tailwind CSS
Backend & Databases: REST APIs, Node.js/Express, PostgreSQL (Supabase), Firebase, Stripe API, PayFast, Git/GitHub
Software Engineering Practices: Software Testing, Object-Oriented Design, Design Patterns
AI & Data: Google Gemini API, NLP, OCR (Google ML Kit), Pandas, NumPy, Scikit-learn, Google Earth Engine, Jupyter Notebook, Google Colab

Certifications

Introduction to Python – DataCamp | Certificate
Intermediate Python – DataCamp | Certificate
Data Manipulation with Pandas – DataCamp | Certificate
Introduction to Object Oriented Programming in Python – DataCamp | Certificate
Intermediate Object Oriented Programming in Python – DataCamp | Certificate